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# Machine learning-based automation of accounting services: An exploratory case study

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ABSTRACT

Machine Learning (ML) applied to Robotic Process Automation (RPA) and chatbot interfaces can generate significant value for many business processes. However, these technologies generate the intended return only with a carefully planned deployment. Current literature only contains a small number of case studies about how the adoption of ML-based automation services impacts employees' behavior. In particular, no case studies look into the automation of manual tasks related to accounting management. This article reports a study conducted to understand users' perceptions of an ML-enabled service to automate repetitive management tasks. The service was developed in a partnership between Unisinos University and Dell Inc. The study was conducted with a group of ten highly skilled employees from Dell with expertise in accounting processes and with IT background that frequently would use the automation service. The group participated in a presentation about the service and its interface and voluntarily answered a Technology Acceptance Model (TAM) questionnaire to evaluate the usability and ease of use. Results show that 10 out of 10 users agree that the service was easy to use. Also, 8 of them agree that its output is useful to reduce the manual labor required for statutory reconciliation. Furthermore, employees with an accounting management background were given access to the service, and three voluntarily answered an open-ended survey. In summary, employees agree that an automation service can reduce the time required to conduct management tasks but questioned the long-term usefulness and the ability to incorporate the process's particularities. These results provided insights leading to ten lessons related to user experience, training and awareness, and service development.

## 1. Introduction

Automation services are a growing trend in a wide range of businesses due to the recent strides of Artificial Intelligence (AI) and Machine Learning (ML) (Soni et al., 2020). This evolution includes the concept of Robotic Process Automation (RPA), which encompasses software services tailored to efficiently execute particular tasks that would require manual and repetitive actions from

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human employees (Syed et al., 2020). Furthermore, chatbot technologies became a popular option for interaction between humans and intelligent software services, both for customers and company employees (Bavaresco et al., 2020). On the one hand, these technologies can generate great value for businesses (Leno et al., 2020). On the other hand, they can disrupt well-known processes and routines, directly impacting the employees' willingness to adopt automation tools (Kulkarni et al., 2019).

Recent studies indicate growing concerns about the influence of automation services on employees and their productivity (CCanhoto and Clearanhoto and Clear, 2020). Employees must have confidence in the changes introduced by these services to avoid negative impacts in the work environment (Berger et al., 2021). To that end, employees must receive proper training to understand the impact of such technologies to achieve the intended return of investment (Lee and Shin, 2020).

Current literature still lacks a broad understanding of how the adoption of ML-based automation services impacts employees' routines (Kokina and Blanchette, 2019; Mirbabaie et al., 2021). Such knowledge enables decisions about what tasks to automate, how to implement the services, and how to promote adoption among potential users. Furthermore, without proper training, employees will end up not fully utilizing these technological resources, resulting in wasted investment without the intended return (Eißer et al., 2020).

This article reports a study conducted to understand users' perceptions of an ML-enabled service to automate repetitive enterprise management tasks. The study employs a deployment of a software service developed in a collaboration between Unisinos University and Dell Inc. This service automates an accounting management task known as Statutory Reconciliation, which presents high repeatability and manual labor. Besides, the service employs a Natural Language Processing (NLP) interface through a chatbot with a natural dialog flow in which users provide the required parameters and receive processing results.

The study was conducted with a group of ten highly skilled employees from Dell Inc. with expertise in accounting processes and with IT background that frequently would use the automation service. The employees participated in a presentation about the service and were invited to answer a series of questions about their experience. The first questionnaire, voluntarily answered by ten participants, employed the Technology Acceptance Model (TAM) (Davis, 1989) to evaluate users' perceptions about how the ML service and NLP interface helped them conduct the accounting task. Later, employees with deep knowledge about the statutory reconciliation process received access to the tool to assess and point out insights. A total of three employees voluntarily answered an in-depth survey about their experience among those who received access. According to collected responses, employees agreed that the automation service and chatbot interface reduced the time required to conduct the statutory reconciliation task. More specifically, TAM results show that 10 out of 10 users agree with the ease of use of the service and interface, while 8 users agree with their usefulness. However, users also questioned the service's long-term usefulness to encompass the automated process's particularities. These results highlight that automation services and respective interfaces require careful design to avoid hindering employees' productivity. Therefore, this article's main contributions are:

- Combine NLP-based interfaces and RPA as a service to support statutory reconciliation, a manual and repetitive task related to accounting management;
- Provide an analysis of how employees perceive the adoption of this ML-enabled automation service;
- Describe ten lessons learned during the evaluation of ML-enabled automation services, encompassing aspects related to user experience, service development, and employee training and awareness.

These contributions highlight critical aspects of planning the introduction of ML-based applications into business teams, aiming to provide guidelines on technology and employee synergy. In addition, the contributions are aligned with previously identified opportunities in accounting, such as selection and evaluation of RPA methods and tools, what types of business processes would benefit from RPA, and what are the possibilities to evaluate RPA (Moffitt et al., 2018).

The remainder of this paper is organized as follows. Section 2 explores other studies related to automation services and employees' perceptions about them. Section 3 explains the methodology adopted to conduct the study. Section 4 presents the results obtained from the user interviews. Section 5 discusses the ten lessons learned based on the results and provides additional insights. Finally, Section 6 presents final considerations and future directions.

## 2. Related work

This section describes studies that focus on *(i)* proposing automation services and chatbot interfaces for management tasks and *(ii)* evaluating the user perception about the usage and impact of these services and interfaces.

A group of studies focused only on proposing intelligent services and interfaces for management tasks. Ramasubbu et al. (2018) explored a system to enable employees to remotely manage their office appliances using a smart plug infrastructure. Godse et al. (2018) automated the ticket system of an enterprise IT department with a chatbot interface that analyzes and proposes solutions to common user problems. Matthies et al. (2019) described a conversational agent tailored to participate on meetings of software development teams and provide metrics and statistics about ongoing projects. Huang and Vasarhelyi (2019) presented a 4-stage framework for applying an automation process in the audit practice. The authors implemented an RPA-based pilot project and evaluated it by comparing the outputs of the system with the manual process. They demonstrated the feasibility of automation in auditing. Androutsopoulou et al. (2019) focused on the automation of governmental services provided to citizens. They proposed a chatbot architecture with advanced expressiveness to provide a digital channel for public service access. Authors demonstrated the framework's use in a series of use cases developed according to results from interviews with government agencies. Rizk et al. (2020) proposed a framework based on multi-agent orchestration for process automation services with chatbot interfaces, focusing on the requirements of enterprise tasks. Authors proposed two case studies to qualitatively demonstrate the performance and scalability of services developed with the framework.

These studies (Ramasubbu et al., 2018; Godse et al., 2018; Matthies et al., 2019; Androutsopoulou et al., 2019; Rizk et al., 2020) focused on designing an intelligent service with a particular goal and a conversational interface that employees can use to interact with it. However, none of them conducted evaluations of the impact on the users' opinions about the service or the impact on their productivity behavior.

Another group of recent studies proposed services with chatbot interfaces that focus on providing tools to help employees' better manage their tasks. Furthermore, they conducted evaluations about the service's impact on productivity and how users perceived its usefulness. Tseng et al. (2019) evaluated a case study on the deployment of a website blocking system with a conversational interface to reduce employee's distractions during work. They argued that the system was effective in reducing distractions without breaking the employee's sense of control over their use of time. Grover et al. (2020) proposed a conversational agent to assist employees in managing time scheduling. Results from a user survey after a three-week experimental deployment indicated that employees felt more productive and increased their time scheduling habits because of the agent. Toxtli et al. (2018) developed an agent that integrates with team management applications and assists with task management. Authors evaluated employee's perception on the usefulness of the bot, which indicates a trend of feeling more productive when using the agent. Sowa and Przegalinska (2020) evaluated the use of domain-specific chatbot assistants to improve the productivity of employees. A survey with the users demonstrated that those who had access to the chatbot had an overall productivity and were more satisfied with their performance.

Additional studies aimed to provide service to automate tasks that are otherwise manually conducted by employees. Some of these studies also employed chatbots interfaces to simplify users' access to services and evaluated the impact and usefulness perceived by users. Piyatumrong et al. (2018) used a chatbot interface to provide employees' with details about companies internal processes and products. They conducted a detailed study in an IT R&D center to evaluate the chatbot effectiveness in reducing employees' information gap about the companies processes. Fiore et al. (2019) evaluated the employee's experience with a chatbot for automation of common IT tasks. Their results indicated that, overall, users appreciated the experience, citing the simplicity and pro-active guidance offered by the chatbot as positive aspects. William and William (2019) proposed a process automation framework to support corporate service providers for small businesses. Their aim was to provide solutions to reduce manual tasks related to documentation generation and review. Also, they deployed a proof of concept service in two service providers in Singapore and verified that the solution greatly reduced the required time and occurrence of errors in the processes results.

The last group of studies focuses on how automation services impact users' perceptions and work routines. Eißer et al. (2020) interviewed employees of a recruiting agency to evaluate their perception on automation anxiety due to the growing usage of chatbots. Results indicated that there is a high level of concern generated by the use of chatbots and the possibility employee replacement with technology. Zhu et al. (2020) conducted surveys to evaluate employee's responses to AI-based automation services. Authors proposed four behavioral profiles to classify employees depending on the willingness to adopt automation services and the chances of leaving the company due to increased changes in their routines. Based on these profiles, the study proposed scenarios to improve the response of

**Table 1**
Related work summary.

| Article | Year | Main Focus | Case Study | Service Deployed | Evaluation with Users | User Impact Analysis |
|---|---|---|---|---|---|---|
| Ramasubbu et al. (2018) | 2018 | Task automation | Appliance automation | ✓ | | |
| Godse et al. (2018) | 2018 | Task automation | IT ticket management | ✓ | | |
| Toxtli et al. (2018) | 2018 | Task management | Employee task management | ✓ | ✓ | ✓ |
| Piyatumrong et al. (2018) | 2018 | Task automation | Enterprise product information | ✓ | ✓ | |
| Matthies et al. (2019) | 2019 | Task automation | Software development metrics | ✓ | | |
| Androutsopoulou et al. (2019) | 2019 | Chatbot architecture | Government services | | | |
| Tseng et al. (2019) | 2019 | Task management | Employee activity focus | ✓ | ✓ | ✓ |
| Fiore et al. (2019) | 2019 | Task automation | IT task automation | ✓ | ✓ | |
| William and William (2019) | 2019 | Automation framework | Business document analysis | ✓ | | |
| Huang and Vasarhelyi (2019) | 2019 | Auditing framework | Auditing task automation | ✓ | | |
| Rizk et al. (2020) | 2020 | Automation framework | Business travel approval | | | |
| Grover et al. (2020) | 2020 | Task management | Employee time scheduling | ✓ | ✓ | ✓ |
| Sowa and Przegalinska (2020) | 2020 | Task management | Employee task management | ✓ | ✓ | ✓ |
| Eißer et al. (2020) | 2020 | Impact on Employees | None | | | ✓ |
| Zhu et al. (2020) | 2020 | Impact on Employees | None | | | ✓ |
| Braganza et al. (2021) | 2020 | Impact on Employees | None | | | ✓ |

employees to automation services. Braganza et al. (2021) focused on employees from technology companies and their response to the anxiety caused by the adoption of AI-based automation services. Results indicated the need to carefully plan the implementation of AI technologies to avoid breaking the long-term trust of employees in the company.

Table 1 summarizes the general focus of the reviewed articles. Most of the articles focused on proposals to automate or manage tasks, including case studies or frameworks. They also described an actual deployment of the proposed service, aiming at evaluating its feasibility. However, not all studies presented an evaluation that includes the perception of users about the service's functionalities. Such an evaluation can be found in 60% of the articles (Ramasubbu et al., 2018; Godse et al., 2018; Matthies et al., 2019; Androutsopoulou et al., 2019; Tseng et al., 2019; Grover et al., 2020; Toxtli et al., 2018; Sowa and Przegalinska, 2020; Piyatumrong et al., 2018; Fiore et al., 2019; William and William, 2019) that deployed a proof of concept. Furthermore, 46% of the surveyed studies (Toxtli et al., 2018; Tseng et al., 2019; Grover et al., 2020; Sowa and Przegalinska, 2020; Eißer et al., 2020; Zhu et al., 2020; Braganza et al., 2021) evaluate how automation services impact the routines and opinions of employees. In particular, this subject is the main focus of three of the most recently published studies (Eißer et al., 2020; Zhu et al., 2020; Braganza et al., 2021). This reflects the growing concern that ML technologies for process automation may deeply impact work environments and the need to manage employee's reception to such changes.

Results in the literature indicate a trend on a good response from users when the service goal is to improve the capabilities to organize their day-to-day tasks. All articles describing such use cases indicate that employees felt more productive while using the proposed service to help them organize their schedule (Tseng et al., 2019; Grover et al., 2020; Toxtli et al., 2018; Sowa and Przegalinska, 2020). In turn, articles describing case studies on services aimed at task automation generated a higher level of mixed responses (Piyatumrong et al., 2018; Fiore et al., 2019). In particular, users with a negative view of the proposed service raised concerns about the ability to deal with particular scenarios of the automated task. Furthermore, they also indicated concerns regarding the substitution of human employees by AI-enabled services.

The literature analysis demonstrates that recent studies show a growing concern regarding the influence of automation services on employees. However, current literature still lacks an empirical analysis of different case studies providing a broad comprehension about how the adoption of ML-based automation services impacts the routines of employees (Zhang, 2019). Particularly, studies that conducted an in-depth analysis on the perspectives of users (Toxtli et al., 2018; Tseng et al., 2019; Grover et al., 2020; Sowa and Przegalinska, 2020) explored services that help employees to organize their tasks, for example to reduce distractions. None of the studies that focus on task automation provided such an analysis. In particular, none of these studies focus on the automation of manual and repetitive tasks related to accounting management. The correct execution of such tasks is critical because of the potential negative consequences of human errors caused by fatigue, distractions, intentional and unintentional errors (Ding et al., 2018). The next section details the accounting task that is subject of this article's case study, as well as the methodology used to evaluate user's perceptions about the developed automation service.

## 3. Methodology

This section presents the methodology adopted in this study. First, it describes the accounting management task selected as case study as well as the proof-of-concept implementation of the automation service and chatbot interface used in the evaluation. Next, it explains the process used to collect the opinions from the users about service and interface on their daily tasks.

### *3.1. Statutory reconciliation automation*

This research aims to evaluate the perception of users about a natural language interface to access an ML-enabled service to aid in the automation of a repetitive management task. The service follows the concept of RPA (Moffitt et al., 2018; Kokina and Blanchette, 2019; Syed et al., 2020). More specifically, it is a software component tailored to perform a given enterprise management process that is well defined, documented, and repeatable.

The service automates a task frequently performed by the accounting management team, defined as "accounting method conversion between US GAAP (Generally Accepted Accounting Principles) and Local GAAP", or simply "statutory reconciliation" (Rueschhoff et al., Jan 1998; Van Schaik, 2021). In summary, the GAAP reconciliation process aims to identify the existing differences when comparing the account balances registered on the local accounting book with the same balances from the corporate accounting book for reporting. These differences can fall into two main categories: timing and permanent differences. More specifically, each reconciliation encompasses the following repetitive tasks executed in a massive number of accounts:

(i) grouping and aggregation of accounts;
(ii) create balances of accounts;
(iii) analyze balances between groups;
(iv) analyze discrepancies in balances;
(v) classify those discrepancies;
(vi) summarize balances based on classifications.

Therefore, the goal of the automation service is to employ a machine-learning algorithm to classify the type of balance difference and calculate its respective value.

Fig. 1 summarizes the workflow for a user to conduct a statutory reconciliation using the automation service. The user accesses the chatbot interface (1) and uploads a spreadsheet with a standard template already known by users, informing the required input parameters (2). In this regard, existing reconciliation processes helped derive both columns and data for the spreadsheet. Next, the automation service receives the input data from the chatbot interface, analyzes its columns, and conducts the required steps to calculate and classify the reconciliation (3, 4, 5). Finally, the chatbot replies to the user (6, 7) with an updated spreadsheet with the obtained results. This automation process can save up to 2 hours from manual labor for each statutory reconciliation. Based on estimates from Dell, 1,500 hours of manual labor can be saved yearly with such automation results.

Employees had access to the statutory reconciliation RPA service through a chatbot interface available in the Microsoft Teams™ communication platform. This platform was chosen based on its availability and widespread usage at the company which participated in the evaluation. In this sense, chatbots potentially facilitate daily tasks because they harness the potential of an already used messaging tool at Dell. Moreover, chatbots have been considered promising to help automate various business processes currently (Bavaresco et al., 2020), since natural language may support processes in a more humanized and flexible way.

The chatbot uses NLP methods to extract the users' intents from dialogues to infer what RPA services and respective parameters are requested. The developed prototype employs the Microsoft LUIS™ NLP framework to analyze users' input and extract their intents. The chatbot also leverages ML to verify and request missing or incorrect parameters to the user prior to submitting them to the RPA service for execution. Furthermore, ML-based history analysis for user dialogues enables the service to predict parameters and requests, further simplifying the interaction.

Fig. 2 illustrates one possible dialogue that invokes the statutory reconciliation service. The dialogue includes the sequences of interactions performed until the upload of a data spreadsheet, which characterizes system input and contains the accounts for the reconciliation process. After the service processes the request, the chatbot interface provides the spreadsheet containing the results of the reconciliation process.

### 3.2. Evaluation process

The evaluation combined a twofold perspective with specific goals in light of developed proof-of-concept: *(i)* the feasibility of using a chatbot as an interface for accounting management automation services; *(ii)* the users' perceptions concerning machine learning employment to automate statutory reconciliation processing. Both perspectives comprised interviews with employees from Dell with business and technical knowledge who could benefit from the features provided by the proof-of-concept service described in Section 3.1.

A total of nineteen employees with expertise in accounting reconciliation procedures at Dell agreed to participate in the evaluation. That is, the population derived from the reconciliation team, and this study did not select specific employees. Therefore, this group participated in a presentation introducing the chatbot interface and the functionalities of the statutory reconciliation service, including a usage tutorial. The group included employees with an IT background and accounting specialists with expertise in statutory reconciliation. Moreover, employees also had mutual knowledge between IT and reconciliation. These employees committed to use the chatbot and system to conduct reconciliation tasks within the company. Later, these employees were invited to voluntarily provide feedback about their experience with the system. A total of ten responses were received to our evaluation.

The questionnaire employed the original version of the Technology Acceptance Model (TAM)(Davis, 1989), which has been considered a standard for this type of evaluation (Marangunic and Granic, 2015; Bavaresco et al., 2020; Barbosa et al., 2018). The model considers two primary focuses. First, it evaluates the *perceived usefulness*, that is, the users' satisfaction and whether the proposed technology can help the users work better. Second, it evaluates the *perceived ease of use*, that is, the user's perceived effort when using a specific system. The results included indicators that identify the feasibility of employees using a chatbot as the interface to

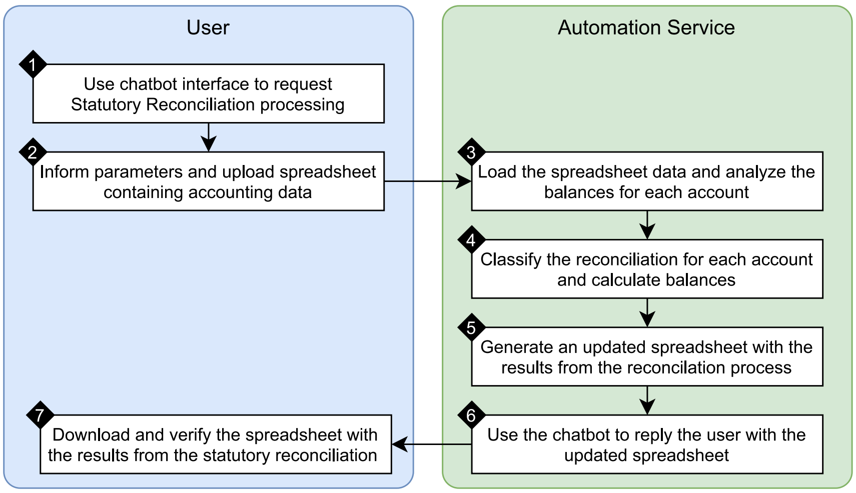


**Fig. 1.** Diagram illustrating the statutory reconciliation process when using the developed automation service.

| Actor | Dialogue |
|---|---|
| Chatbot | Welcome to the Accounting Management Bot!<br>You can interact with me for various accounting automation processes. If you would like to know more about how I can help you, read my documentation by clicking the button below.<br>How can I help you today? |
| Employee | Please provide a statutory reconciliation. |
| Chatbot | Got it. I am going to guide you through a Statutory Reconciliation.<br>Please, what is the Legal Entity and Financial Year End? |
| Employee | The legal entity is 7070 and the financial year end is January 2018. |
| Chatbot | And, what is the origin accounting method and destination accounting method? |
| Employee | From parent to subsidiary. |
| Chatbot | Ok! I will proceed for Legal Entity: 7070, Financial Year End: 2018-01-01, From: parent to subsidiary.<br>Please, upload a XLSX spreadsheet with the data for Reconciliation. |

**Fig. 2.** Sample dialogue from the chatbot interface.

automate accounting management.

Two major categories divided the questions according to the objectives of the TAM, where each category included three questions. The responses were classified using a Likert scale with five points, namely, *Strongly agree, Agree, Neither agree nor disagree, Disagree, Strongly disagree* (Likert, 1932). Additionally, one last descriptive question allowed employees to express themselves freely about their perceptions. Table 2 presents the TAM questionnaire.

In the final step of the evaluation, employees with a business background who would use the automation service as a tool for daily tasks had access to the proof-of-concept via the chatbot deployed into Microsoft Teams™. These employees belonged to the initial group, and they were required to use the service in real situations for one week. They performed interactions with the chatbot, and they submitted a list of accounts using the spreadsheet for statutory reconciliation processing. After the evaluation period, the users received a two-question survey with discursive questions to voluntarily expose their experience, as shown in Table 3. A total of three users provided detailed feedback, allowing to retrieve specific and personal points of the chatbot in addition to reconciliation processing.

## 4. Results

This section presents the results obtained during the evaluation according to the proposed methodology, beginning with the TAM assessment results. Fig. 3 depicts the results from the six questions contained in the TAM questionnaire, as listed in Table 2.

Regarding ease of use, results from Q1 show that 8 employees strongly agreed that the chatbot's dialog interactions were easy to understand. The remaining 2 employees also agreed with the sentence but indicated that there is room to clarify the information conveyed with the chatbot's dialogues. Answers to Q2 indicated that 6 employees strongly agreed and 4 agreed that the sequence of dialogues used in the chatbot was consistent with terminology from the accounting business area. However, the open question responses indicated the need to expand the chatbot's vocabulary by improving the natural language processing model. In particular, employees pointed to the need to diversify the vocabulary and support other idioms for international usage. Results from Q3 present that 5 employees strongly agreed and 5 agreed that the results on the spreadsheet after the machine learning processing were clear and easy to understand. Nevertheless, users mentioned doubts regarding the need to adapt the spreadsheet to local business characteristics, even considering that the template was similar to that used in their day-to-day manual activities. This observation stemmed from the need to adjust the spreadsheet template employed by employees for the reconciliation task, deviating from the traditional format that employees were trained to use.

In the subject of usefulness, answers to Q5 and Q6 presented a similar behavior, with a strong agreement of 6 employees and

**Table 2**
Questions applied with the TAM methodology.

| | |
|---|---|
| **Ease of use** | |
| **Q1** | The sequence of dialog interactions provided by the chatbot is easy to understand. |
| **Q2** | The dialogue presented is consistent with the accounting business area concerning the terms used. |
| **Q3** | The results presented in the spreadsheet after processing are clear and easy to understand. |
| **Usefulness** | |
| **Q4** | The chatbot can understand and respond to a dialogue while maintaining the minimum expected quality according to the proposed scenario. |
| **Q5** | The generated spreadsheet can improve the productivity of users who carry out statutory reconciliation. |
| **Q6** | I would use the output of the service to assist me in the accounting reconciliation process. |

**Table 3**
Discursive questions (DQ) for users to offer additional insights on their experience.

| | |
|---|---|
| **DQ1** | What are the easiest and hardest aspects during usage of the tool? |
| **DQ2** | Would the tool improve the results of your job? Why? |

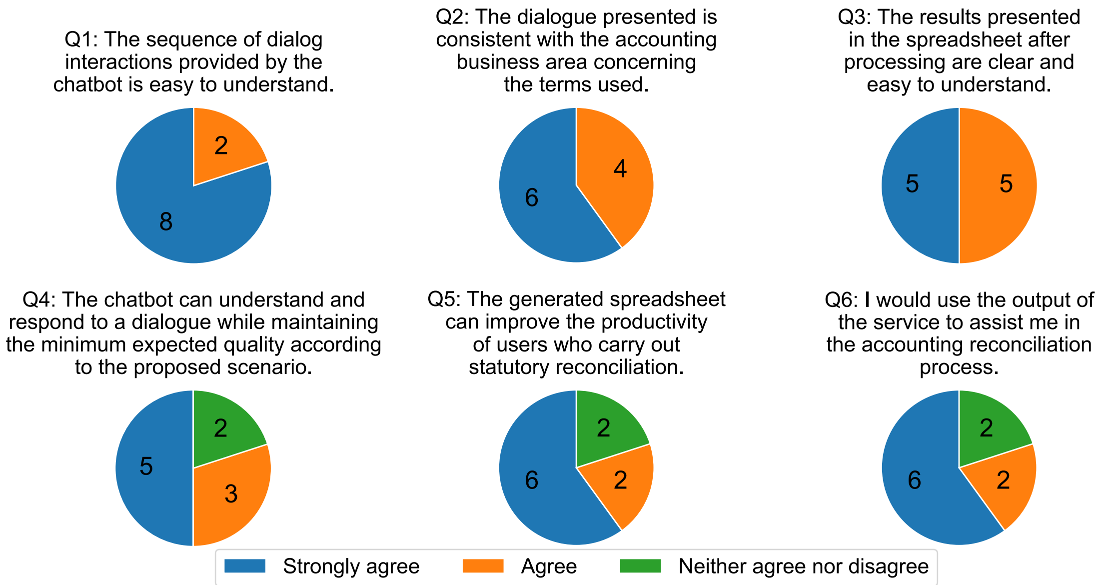


**Fig. 3.** Detailed results from the TAM evaluation.

agreement of 2 employees. First, employees indicated that the spreadsheet processed with the machine learning process could improve statutory reconciliation activities' productivity. Second, employees would use the spreadsheet if they had access to it in their daily activities. Another group of 2 employees had a similar, neutral position on both questions. This group pointed out the need to customize the spreadsheet to local demands and observe its use in real scenarios.

Finally, results from Q4 indicated that 5 employees strongly agreed and 3 agreed that the chatbot achieved a minimum expected quality level on the understanding and responses to user interactions. The remaining 2 employees remained neutral to the statement. These comments indicated that dialogues should consider the characteristics of the accounting usage scenario in more depth and observe the tool's day-to-day usage by employees in natural environments.

Fig. 4 illustrates the summary of TAM results considering ease of use and usefulness. It shows 63% of strong agreement and 37% of agreement from users regarding the ease of use of the prototype. In turn, there is 57% of strong agreement and 23% of agreement from users about the usefulness of the service and its results. These results indicate an overall satisfaction from employees when interacting with the RPA service, which also has been demonstrated in literature (Cooper et al., 2019). However, 20% remained neutral regarding the service's usability. This result is in line with additional comments from employees that pointed out aspects requiring improvements before deployment for broader usage. In particular, they mentioned the need to expand the chatbot's vocabulary to accommodate multi-region scenarios and the spreadsheet template features. These suggestions appeared in the open answers accompanying the questionnaire. One employee mentioned the *"(…) opportunity to evaluate for other regions and other accounting scenarios (…)"*. Another suggested that *"(…) the solution is adequate, requiring only adaptation to local scenarios (…)."* The responses indicated the negative aspects raised by users regarding the dialog models used in the chatbot. To improve these aspects, the NLP learning models require

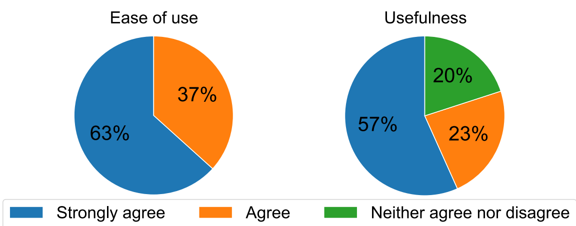


**Fig. 4.** Summary of the TAM results considering the evaluation aspects of ease of use and usefulness considering the response from 10 users. Each graph considers the average values for all questions encompassing a given aspect. In summary, 10 out of 10 users agreed with the ease of use of the service, while 8 of them with the usefulness of the results. Furthermore, no users disagreed with both aspects.

further training with expanded dialogue datasets.

The second part of the results relates to the two-question survey conducted after the chatbot's usage in a real scenario. Three answers provide an in-depth understanding of the employee's perspective on the chatbot and service usage. They present the perspective of users with a business background who would use the chatbot as a tool for daily tasks. Fig. 5 depicts the complete answers provided by users of the discursive questions, showing their perspective on the offered experience. The comments evidence previous results related to usability and business requirements. The first question overviews aspects of chatbot interaction and language understanding. In turn, the second highlights how the service fits into the daily tasks of accounting operations. Hence, the latter primarily considers machine learning processing. In summary, the answers show that users agreed that the tool would be useful for expediting their daily tasks. However, they also indicated concerns about the service's long-term usefulness, given the particularities of the automated task. Further analysis shows that such concerns occur because users expected a "production quality" software while the experiment used a proof-of-concept that still requires refinement for wide availability.

## 5. Discussion

This section discusses the case study's implications from industrial and academic perspectives. Furthermore, it highlights ten lessons learned throughout this research. From an industrial perspective, the study presents results that guide the development and deployment of automation services. Furthermore, this research analyzes a case study regarding the development and deployment of an automation service to reduce employees' demands on a repetitive accounting management task. Besides the savings in employee hours due to automation, the service may reduce human errors caused by repetitive tasks, which has been demonstrated valuable for the accounting area (Moffitt et al., 2018; Zhang, 2019). Hence, the service also reduces the chances of financial loss due to errors in tax reports, considering that financial operations must receive careful and thorough executions.

In particular, results provide insights into aspects that may cause employees to mistrust a technology due to fears of disruption and uncertainty in their routines and jobs. This is also discussed in implications for accountants (Kokina and Davenport, 2017; Cooper et al., 2022) and observed in previously interviews (Kokina and Blanchette, 2019). The responses to the second question of the open-ended survey mainly guided these insights, which present opposition of current functionalities, claims about proposed solutions, and demotivation of future developments. Respectively, this is depicted by.

(i) user 3, who mentioned manual operations on the spreadsheet instead of system usage;
(ii) user 2, who mentioned training of a new tool; and
(iii) user 1, who wrote about future difficulties for the system to learn and implement account adjustments

In this sense, the insights are essential to reduce possible frictions and increase investment return in automation services development. Therefore, these aspects led to the discussion of lessons learned concerning the employees' approximation of potential novel

| User | Question | Answer |
|---|---|---|
| User 1 | DQ1 | "The hardest part is that the tool was not able to detect our request and responded 'Sorry, try asking in a different way'. That means users have to memorize the standard wording to get it to work. Besides, as of now, we are not able to verify if the output of the reconciliation file is produced as expected." |
| | DQ2 | "I think the time consumed when using the tool is longer compared to manually performing the task. It takes time to chat with the accounting bot and prepare the template to upload to the tool. As every year entities might have new accounts and audit adjustments, it will be hard for the tool to learn how to split into permanent and timing difference." |
| User 2 | DQ1 | "Drawbacks: - Chatbot only recognized standard response from the user. Advantages: - Easy accessible on the completed reconciliation (from share point)" |
| | DQ2 | "Not really. *(i)* Not user friendly - Chatbot only recognized standard responses from the user. New users have to be trained to use the BOT. *(ii)* Time taken for BOT to run the reconciliation - I did the testing and it took BOT to get back to me on the completed reconciliation after 1 hour. *(iii)* Data validation - manual validation required to be done to ensure completeness and accuracy of data compiled by BOT. *(iv)* Functionality limitation - the BOT must be able to cope with addition of new accounts, new types of audit adjustments, foreign currency issues, specific local statutory requirements that are relevant to each of the entities, changes of reconciliation template in future." |
| User 3 | DQ1 | "Hardest: May not get response when using actual file; Easier: Communication." |
| | DQ2 | "Limited for statutory reconciliation. As the only two functions are to *(i)* identify whether the difference permanent or timing, which will be classified when preparing difference list, and *(ii)* summarize values from supporting tabs to cover page, which would just take several minutes by using advanced functions from the spreadsheet." |

**Fig. 5.** Answers to the discursive questions provided by three Dell's employees.

tools, participation in the development process, and explanation of the impact of new technologies.

From an academic perspective, the study's results contribute to the body of knowledge related to the impact of ML and automation technologies in the industry. The results provide insights into employees' perceptions about an automation service for a repetitive task for accounting management. These results include in-depth responses, including the advantages and limitations that indicate the most relevant aspects to the service's users. These concerns encompass a relevant area of study, gaining attention due to the popularization of automation services and the growing concerns regarding the disruption they will cause in nearly all industry fields. The related work presented in Section 2 evidences this relevance. In particular, recent works (Eißer et al., 2020; Zhu et al., 2020; Braganza et al., 2021) explore the impact of automation technologies on employees' behavior and how the lack of proper support can lead to "automation anxiety" in the workplace. Conversely, other studies (Rizk et al., 2020; Grover et al., 2020; Sowa and Przegalinska, 2020) present results indicating that employees can leverage automation services and feel more productive if they understand the purpose and impact of these tools in their work routine. In that sense, this study's results present a practical example of integrating employees in automation services development.

Fig. 6 depicts ten lessons learned from this research. These lessons provide technology and employee synergy guidelines in light of machine learning-enabled services. They encompass three different perspectives organized by the context they apply, depicted by user experience, training and awareness, and service development.

| Context | Lesson | Description |
|---|---|---|
| User experience | Employees have high expectations from the chatbot interface | Results show that employees had high expectations regarding the chatbot capacity to recognize dialogues. This is highlighted in responses indicating that the chatbot should have its vocabulary expanded, both in regard to accounting terminology and localization support. |
| | Employees expect the interface to recognize terminology specific to their field | Employees indicated that the chatbot needs support for additional vocabulary that may be encountered in daily usage for statutory reconciliation. This observation reflects the general expectation from users that they would evaluate a "production level" software instead of a proof-of-concept. |
| | Employees were willing to adopt the automation service to improve their day-to-day activities | Employees demonstrated an overall positive response to the service's results, indicating interest in having it available as a production tool for daily usage. This interest also justifies comments related to the quality of dialogues and automation service. |
| Training and awareness | Employees need a thorough understanding of the contributions of ML to the work routine | Employees must be aware of the possibilities and benefits brought by the ML technologies of automation services. Such knowledge helps reducing the "friction" caused by this technology in employees' routines. Furthermore, employees will fell more compelled to share their particular knowledge on management processes, leading to identification of opportunities for improving automation tools. |
| | Develop a continuous program to present and explain employees the impact of new technologies inserted in their workspace. | Employees tend to avoid adopting technologies when they do not understand their functionalities, even when there are well-known benefits such as the reduction of time spent on tasks. Thus, there is the need of additional efforts to mitigate adverse impacts on users' perception, such as the feeling of being surpassed by technology and the lack of sensibility regarding human resources. |
| | IT personnel needs proper training to manage implementation of ML-based automation services | AI and ML-based techniques require properly curated data sources to achieve high quality results on automation services. IT professionals and management need to be aware of the capabilities and limitations of ML-enabled technologies. Such awareness is key to "demystify" these technologies and establish a common understanding about the results achievable with them, which enables a realistic evaluation of the applicability of automation services. |
| Service development | The automation service requires global input to achieve the expected accuracy | Employees indicated that the service's output may require adaptions to be compatible with requirements and regulations from different countries. This observation suggests that users expected to interact with a "production level" software ready for use in a global level. While this is a fair assumption from users, such level of functionality was not intended in the initial prototype. |
| | The automation service must be adaptable to changes in accounting practices | Employees mentioned that the current automation service seems to be tailored to current accounting standards regarding statutory reconciliation. They indicated that the service must be capable of adapting to changes that occur routinely in accounting regulations and practices. Hence, this process of adaptation must be properly evaluated to identify the need of human interaction when needed. |
| | Employees should participate in the development process of the service and interface | Employees that participate in the development process have a better understanding of the functionalities and how to integrate the automation service in the work routine, reducing potential negative opinions. Furthermore, early input from potential users enables a better adaption of the service to the particularities of the business process, given the expert knowledge they can provide. |
| | Request service users to provide dialogue inputs for the service's chatbot interface | Employees who have daily contact with the automated processes have a deep knowledge of terminologies and nomenclatures. Such knowledge is essential to train the chatbot because it leads to dialogue sequences more natural to employees. As a result, dialogues become more streamlined, which minimizes the time required for users to adapt to the interface. |

**Fig. 6.** Lessons learned during the conducted evaluation, highlighting key aspects that impact the adoption of automation services. These lessons pertain three major aspects: user experience, training and awareness, service development.

Regarding the user experience lessons, responses to questionnaires indicated a high expectation from the chatbot interface and automation service features. This became evident in comments mentioning that the automation service must support more accounting scenarios or the chatbot needs to understand a wider vocabulary used in daily accounting tasks. Such comments indicate that employees expected to receive a “polished” tool that they could integrate into their workflow to improve their productivity instead of software that would force an adaption of their work routine. In this sense, the development process needs to be aware of users’ expectations and enterprise processes workflow to introduce a tool that effectively improves productivity and does not become a burden for users, which leads to lessons learned in the service development context. This aspect is particularly important to promote a positive opinion on AI-based automation services given their potential negative reputation.

Besides, users expected the chatbot interactions to be as natural and streamlined as possible, including support for the business process’s vocabulary. This fact resulted from users expecting the adoption of a new tool to cause the smallest possible disruption in their daily routines. This revealed they expected to interact with the chatbot interface as if they were conversing with a highly trained professional in the automated task. Such a requirement adds to the simplicity and ease of use that an NLP interface should offer users.

Evaluation results also indicated that employees were willing to adopt the automation service and chatbot interface to improve productivity by reducing hours spent with menial tasks. However, users indicated they expected to use a service capable of providing results on par with those obtained if they dedicated their time manually executing the task. On the one hand, this demonstrated the need to consider business employee input during service development to guarantee that the resulting product met such expectations. Nevertheless, this also indicated the need to communicate with employees to manage their expectations and avoid an initial negative opinion on adopting automation services. This is essential due to potential misconceptions about AI-enabled technologies and their consequences to employees’ routines.

Concerning training and awareness, results indicated that employees need a thorough understanding of ML’s contributions to their routines. In particular, employees must know the benefits that automation services can bring, for example, reducing menial tasks that can lead to errors due to fatigue. As a result, they can focus efforts on other tasks that result in higher productivity. From a broader perspective, providing employees with knowledge about the automation services’ scope and goals mitigates negative opinions due to conflicting information. Such efforts help mitigate misinformation created by “popular knowledge” such as that ML-based automation will reduce employment. If such misinformation persists, employees will not develop the necessary trust in these services to use them effectively.

Besides employees, enterprise management also needs to understand automation services, particularly their capabilities and limitations. Technology advancements publicized in the media may give false impressions regarding the types of tasks that can be automated and the complexity involved in their development. ML-based automation services require large datasets and detailed input from experts to encompass all the aspects related to enterprise processes. Thus, it is essential to provide a thorough understanding of these technologies to enterprise management to avoid creating false expectations about the types of tasks that can be automated. Such knowledge also helps the information about the automation services to spread within the organization’s hierarchy, reducing misinformation.

Concerning development, an automation service aimed at multinational companies must consider the particularities imposed by various locations. This requirement appeared in answers to the evaluation questionnaire, indicating that the current proof-of-concept is potentially over-fitted to scenarios from particular regions. Moreover, future automation may consider regulatory requirements relevant to fit legal aspects that frequently present changes. In this case, further development of the prototype requires collecting additional data to cover a broader set of requirements to achieve a wider audience within the company. If such a requirement is overlooked, the service may burden regions with particularities overlooked during implementation, leading to a poor overall opinion of users and automation services. Hence, these aspects reveal potential actions that companies may consider aligned with the presented lessons. In particular, employees should participate in the development process with business and system analysts to facilitate requirements gathering, communication, and understanding of the proposed software solution.

Another factor that appeared in users’ requirements is the adaptability of the service to changes in the workflow of the automated task. The evaluation presented in Section 4 shows that users observed the need to guarantee that the service is compatible with different accounting practices and guidelines, including future changes. A service aimed at reducing the burden on employees and increasing their productivity cannot lose this property due to changes in the workflow it is intended to automate. Thus, the design must include the means to update and maintain the ease of use intended in its conception. As an example, ML-enabled services can use feedback and continuous learning methods to adapt to changes in activities. This means that employees’ suggestions and corrections can be introduced in further implementation tasks, particularly in identifying novel data standards to retrain ML services. Hence, these methods enable continuous improvement in service quality for automated processes.

Concerning continuous improvements, the recommendations for following research regard understanding the differences and similarities between short-term and long-term projects and evaluations. For example, in a long-term RPA project, would the accountants be willing to improve and contribute to the RPA and machine learning services along with data scientists and developers? Would these employees be motivated by the new functionalities of the machine learning services? Would employees have fewer expectations of dialogues once they participated in the whole system’s enhancements? In addition, this work provides questions to be addressed based on long-term evaluations. Would this work’s results and current related researches have the same outcome after long-term usage of RPA? Would employees evolve professionally due to constant training, education, and contact with other areas? In this sense, do companies have new roles for ongoing RPA projects? Finally, would these employees have increased productivity, and what is the return on investment of the long-term RPA usage? These are some of the opportunities from unanswered questions constructed on the limitations of this work.

## 6. Conclusion

Robotic process automation (RPA) and natural language processing (NLP)-enabled interfaces are growing in popularity as tools to improve productivity and offer a more streamlined interaction with clients. These technologies can bring considerable disruption to employees' workflows in daily tasks, consequently generating insecurities. Understanding employees' expectations about automation services is crucial to reduce negative impacts and foster adoption. This paper studied users' opinions regarding a proof-of-concept ML-enabled automation service with a chatbot interface to contribute to this goal. In particular, the RPA service aimed to automate the statutory reconciliation task, which is a manual process with a high potential for automation.

The evaluation included a questionnaire based on the TAM and open questions aimed at understanding how users perceived the RPA service and respective chatbot interface. Results indicated that users had an overall good opinion about the service and its potential to improve productivity by reducing the manual steps to complete the automated task. More specifically, TAM results show that 10 out of 10 users agree with the ease of use of the service and interface, while 8 users agree with their usefulness. Furthermore, users provided a series of observations related to their expectations on how the service would adapt to real usage scenarios under different circumstances. This leads to ten lessons about adopting automation services, encompassing aspects related to user experience, service development, and employee training and awareness.

Therefore, this study noticed that business users are interested in improving their productivity and reducing time-consuming tasks through novel automation services. However, they also expect a service that will seamlessly integrate into the routines and not become a burden during their work. That is, companies should embrace novel technologies without forgetting that employees desire smooth experiences in front of already conducted procedures. This potentially reveals that users can be strict about working changes, and using a familiar and already company-adopted tool favors technology and employee synergy, such as harnessing a communication platform to automate services using natural language. However, in this specific practice, employees will expect high-quality dialogue utterances. Finally, the study fosters careful design and employment of machine learning methods supported by a clear explanation of possibilities to employees.

Future work of this study includes further development of the prototype service, such as eliminating the spreadsheet as incoming data. Hence, additional validation of errors and noise in data becomes necessary in future changes. Moreover, the dialogue interface from the chatbot can be trained with different terminology as suggested by the employees that participated in the evaluation. This also encompasses the opportunity to embrace other accounting fields toward the chatbot being a significant entry-point for all Dell's accounting procedures.

Further, studies with a broader audience with limited technological skills can provide additional insights and comparisons about NLP-enabled interfaces and RPA in light of the limitations of this study. In this sense, the main limitation of this work regards the survey's number of respondents and the generalization of results. Future works may employ metadata studies on the literature for long-term analysis and comparison toward the abovementioned generalization. Besides, long-term research may benefit the understanding of employee evolution due to the technological support, in addition to enhancements in interfaces, dialogues, machine learning models, and the return of investment of these artifacts. This means that following a more significant number of employees and their insights for long-term behavior identification in different accounting processes, managerial levels, skill sets, and profiles corroborates addressing the current limitations of this work.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.accinf.2023.100618, and at https://researchbox.org/1294.

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